

Grade 8
Unit 3
Lesson 4 Practice

Name _____
Date _____
Period _____

1. Consider the value 4.8×10^{-6} . Complete the statements. Then express this value in standard form.

10^{-6} relates to the _____ place. When rewriting 4.8×10^{-6} in standard form, the ____ will be in the millionths place, followed by the _____. The places before the 4 will each hold the digit, ____.

2. The diameter of an atom in gold is about 1.44×10^{-10} meters. Express this value as a decimal in standard form.
3. The average mass of one ant is about 2.9×10^{-6} kilograms. Express this value in standard form.

Express each number in scientific notation.

4. The diameter of a human hair is about 67 hundred-thousandths of an inch.
5. 0.0000000049

Express each number in standard form.

6. A cell's diameter 0.0174×10^{-4}
7. 9×10^{-6}

8. Complete the table.

Scientific Notation	Standard Form
	0.000000026
5.2×10^{-3}	

9. Complete the statements using the table in question 8.

○ 5.2 is ____ times

☐ smaller
☐ larger

 than ____.

○ 10^{-3} is ____ times

☐ smaller
☐ larger

 than 10^{--} .

○ The value 5.2×10^{-3} is ____ times

☐ smaller
☐ larger

 than 0.000000026.

10. Match each value on the left with its expression in scientific notation on the right by drawing lines to connect the equivalent expressions.

0.000144

$1.44 \cdot 10^{-7}$

0.0000144

$1.44 \cdot 10^{-2}$

144 hundredths

$1.44 \cdot 10^{-4}$

0.00144

$1.44 \cdot 10^{-3}$

144 ten millionths

$1.44 \cdot 10^{-5}$